

Aziza Zhanabatyrova

Robotics and computer vision engineer, Aalto University

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in Aziza Zhanabatyrova

📍 Helsinki, Finland

Summary

PhD researcher with 6+ years of international experience in computer vision, robotics, and applied AI. Specialized in SLAM, 3D reconstruction, and deep learning for autonomous systems. Proficient in Python, PyTorch, and TensorFlow, with a track record of designing advanced neural architectures for real-world perception tasks.

Work experience

- 2025 – 2026 **Perception and Autonomy Lead** | Hertel robotics, Lisbon, Portugal
Contributed to early-stage research and planning for a UAV project. Focused on perception and autonomy concepts and initial system design. Supported foundational project organization and technical preparation.
Topics: *Robotics, Computer Vision, Perception*
- 2024 – 2025 **Machine learning engineer** | Pompeu Fabra University, Barcelona, Spain (6 months)
Created and deployed a visual positioning system using deep learning and computer vision. Transitioned from visual-inertial odometry (VIO) to absolute relocalization, reducing drift accumulation.
Topics: *Deep Learning, Computer Vision, 3D mapping, Camera re-localization*
- 2023 – 2024 **Visiting researcher** | National Institute of Informatics, Tokyo, Japan
Conducted research on robust visual change detection.
- 2019 – 2024 **Doctoral researcher** | Aalto University, Finland.
Focus: 3D mapping and localization for autonomous driving.
Topics: *Deep Learning, Computer Vision, 3D mapping, Localization, Crowdsourcing, Object detection, Structure from motion, Change detection: visual and probabilistic.*
- 2018 – 2019 **Research assistant** | Aalto University, Finland
Worked on 3D mapping and positioning, including SfM, image-based relocalization, object detection, semantic segmentation, depth estimation.
- 2015 – 2016 **Computer science intern** | Lead Creation, Sydney, Australia
- 2014 – 2015 **Lab assistant in C, C++, PHP** | EMU, North Cyprus
- 2013 – 2014 **Computer science intern** | Kaznetmedia, Astana, Kazakhstan

Education

- 2019 – 2024 **Ph.D. in Deep Learning** | Mobile Cloud Computing lab, Aalto University, Finland. G.P.A.: 4/4
Thesis: *Crowdsourced 3D semantic mapping and change detection in urban driving environments.*
Topics: *Deep Learning, Computer Vision, 3D mapping, Localization, Crowdsourcing*
- 2017 – 2018 **Master thesis in Computer Vision** | ETH Zurich University, Switzerland.
Thesis: *Domain adaptation using GANs for semantic segmentation of urban scenes.* G.P.A.: 3.6/4.
Topics: *Deep Learning, Domain adaptation, Semantic segmentation*
- M.Sc. Robotics Engineering** | Jaume I University, Spain. G.P.A.: 3.44/4.
- 2016 – 2017 **M.Sc. in Robotics Engineering** | University of Genoa, Italy. (1st year)
- 2014 – 2015 **B.Sc exchange in Computer Science** | KoreaTech, Cheonan, South Korea. G.P.A.: 4/4.
- 2010 – 2015 **B.Sc. in Computer Science** | Eastern Mediterranean University, North Cyprus. G.P.A.: 3.6/4.0.

Skills

- Languages **English (Fluent, 5/5), Russian (Native, 5/5)**
- Coding **Python, C, C++**
- Frameworks **TensorFlow, PyTorch**
- Libraries **OpenCV, scikit-learn, NumPy, SciPy, Pandas, Transformers**
- Tools **ROS, Docker, Kubernetes, Git, AWS, TensorBoard, Jupyter**

Skills (continued)

Topics	■ SLAM, SfM, 3D mapping, camera re-localization, change detection, semantic segmentation, object detection, depth estimation
Analytic	■ Experimental design and evaluation, cross-modal data analysis, real-world problem solving, visual analytics for sensor and image data

Miscellaneous Experience

2024 ■ Authored funding proposals to secure financial support for research projects | Aalto University, Finland

Research Advisor

- 2025 ■ Chauhan Inshu, PhD: "Change detection in construction sites using spherical videos" | Aalto University
- 2021 ■ Rafael Pires Orozco, Master thesis: "Object detection in urban scenes" | Aalto University, Finland
- 2020 ■ Krishna Karthikeyan, Master thesis: "QoI-oriented incentive mechanisms" | Aalto University, Finland

Awards and Achievements

- 2023 ■ **Funding for research visit abroad** | HIIT, Finland.
- **NII research visit scholarship** | NII, Tokyo, Japan.
- 2018 ■ **Four-year phd scholarship** | Aalto University, Finland.
- 2017 ■ **Erasmus scholarship** | University Jaume I, Spain.
- 2016 ■ **Emaro+ consortium scholarship** | UNIGE, Italy.
- 2015 ■ **KGSP scholarship** | NIIED, South Korea.
- 2011 ■ **Four-year bachelor's scholarship** | EMU, North Cyprus

References

Yu Xiao, Associate Professor at Aalto University, Supervisor at Aalto University and Co-author, yu.xiao@aalto.fi
Clayton Leite, Postdoctoral Researcher at Aalto University and Co-author, clayton.souzaleite@aalto.fi

Research Publications

Journal Articles

- 1 **Aziza Zhanabatyrova**, C.F.S. Leite, and Yu Xiao. "Automatic Map Update Using Dashcam Videos". In: *IEEE Internet of Things Journal* (2023), pp. 1–19. [DOI: 10.1109/JIOT.2023.3244693](https://doi.org/10.1109/JIOT.2023.3244693).
- 2 **Aziza Zhanabatyrova**, C.F.S. Leite, and Yu Xiao. "Structure from Motion-Based Mapping for Autonomous Driving: Practice and Experience". In: *ACM Trans. Internet Things* 5.1 (Jan. 2024). [DOI: 10.1145/3631533](https://doi.org/10.1145/3631533).
- 3 **Aziza Zhanabatyrova** et al. "Object Localization and Change Detection in Urban Environments Using Dashcam Videos". In: *ISPRS Archives XLVIII-1/W5-2025* (Nov. 2025). Presented at the 13th International Conference on Mobile Mapping Technology (MMT 2025), Xiamen, China, June 2025, pp. 169–176. [DOI: 10.5194/isprs-archives-XLVIII-1-W5-2025-169-2025](https://doi.org/10.5194/isprs-archives-XLVIII-1-W5-2025-169-2025).

Conference Proceedings

- 1 Giancarlo Pastor et al. "Qualifying 5G SA for L4 Automated Vehicles in a Multi-PLMN Experimental Testbed". In: *2021 IEEE 93rd Vehicular Technology Conference (VTC2021-Spring)*. 2021, pp. 1–3. [DOI: 10.1109/VTC2021-Spring51267.2021.9448788](https://doi.org/10.1109/VTC2021-Spring51267.2021.9448788).
- 2 A. Zhanabatyrova, C.F.S. Leite, and Y. Xiao. "Detecting and Classifying Changes in Traffic Rules using Induction Loop Data". In: *2023 IEEE International Conference on Big Data (BigData)*. Los Alamitos, CA, USA: IEEE Computer Society, Dec. 2023, pp. 1248–1255. [DOI: 10.1109/BigData59044.2023.10386419](https://doi.org/10.1109/BigData59044.2023.10386419).

Under Review/ In Progress

- 1 I. Chauhan, A. Zhanabatyrova, and O. Seppanen. *Change Detection in Construction sites using Spherical Videos*.
- 2 C.F.S. Leite et al. *Transformer-Based Approaches for Sensor-Based Human Activity Recognition: Opportunities and Challenges*.